

1. SYNCHRONIZATION (Process Synchronization)

Simple Meaning:

Synchronization ka matlab hai multiple processes ko aise control karna ki jab wo shared resource use karein to data corrupt na ho.

Technical Definition (Interview):

Synchronization is the mechanism in operating systems that coordinates the execution of concurrent processes to ensure that shared resources are accessed in a controlled manner, maintaining data consistency and preventing race conditions.

Need of Synchronization:

- Multiple processes run concurrently
 - Shared resources (memory, files, variables)
 - Without control → inconsistent output
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Race Condition:

A situation where multiple processes access and modify shared data simultaneously and final result depends on execution order.

Example:

Bank balance = 5000

P1 withdraw 3000

P2 withdraw 3000

 wrong final balance possible without sync

Critical Section:

Part of code where shared resources are accessed.

Structure:

1. Entry Section
 2. Critical Section
 3. Exit Section
 4. Remainder Section
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Critical Section Problem Requirements:

1. Mutual Exclusion

Sirf 1 process CS me ho

2. Progress

Agar CS empty hai to selection delay na ho

3. Bounded Waiting

Har process ko fair chance mile

Synchronization Tools:

1. Mutex (Lock)

- Binary lock (0/1)
- Only one process allowed in CS

2. Semaphore

- Integer variable
- Types:
 - Binary Semaphore (0/1)
 - Counting Semaphore (0 to n)
- Operations: wait() and signal()

3. Monitor

- High-level synchronization
 - Automatic mutual exclusion
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Real Life Example:

Washroom lock system → only one person allowed

Interview Questions:

- What is synchronization?
 - What is race condition?
 - Difference between mutex and semaphore
 - What is critical section?
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2. DEADLOCK

◆ Simple Meaning:

Deadlock tab hota hai jab processes ek dusre ke resources ka wait karte rehte hain aur system permanently ruk jata hai.

💡 Technical Definition (Interview):

Deadlock is a situation in an operating system where a set of processes are permanently blocked because each process is holding at least one resource and waiting for another resource held by another process in the same set.

🔥 Example:

- P1 → Printer hold, Scanner wait
 - P2 → Scanner hold, Printer wait
👉 both stuck forever
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🚫 Coffman Conditions:

Deadlock tab hota hai jab ye 4 conditions simultaneously exist kare:

1. Mutual Exclusion

Ek resource ek time pe ek process use kare

2. Hold and Wait

Process ek resource hold karke dusre ka wait kare

3. No Preemption

Resource forcibly nahi le sakte

4. Circular Wait

Processes circular chain form karein

🧠 Deadlock Handling:

1. Prevention

Conditions ko break karna

2. Avoidance

Safe state maintain karna (Banker's Algorithm)

3. Detection

Cycle detect karke recovery

4. Recovery

Process kill ya resource reclaim

Banker's Algorithm:

- Resource allocation safe hai ya nahi check karta hai
 - System ko unsafe state me jane se rokta hai
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Interview Questions:

- What is deadlock?
 - Explain Coffman conditions
 - Banker's algorithm
 - Deadlock vs starvation
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3. MEMORY MANAGEMENT

Simple Meaning:

Memory management OS ka kaam hai RAM ko processes ke beech efficiently distribute karna.

Technical Definition (Interview):

Memory management is the process of controlling and coordinating computer memory, including allocation, deallocation, protection, and optimization of memory usage for multiple processes to ensure efficient system performance.

Need:

- Limited RAM
 - Multiple processes run simultaneously
 - Efficient utilization required
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Functions:

- Allocation of memory

- Deallocation of memory
 - Protection of memory
 - Sharing of memory
 - Address translation
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1. MEMORY MANAGEMENT TECHNIQUES (MAIN TYPES)

1. Contiguous Allocation

- Process ko continuous memory block milta hai
- Simple but inefficient

Types:

- Fixed Partitioning
- Variable Partitioning

Problems:

- Fragmentation
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2. Non-Contiguous Allocation

(A) Paging ★

- Memory fixed size blocks me divided
- Pages (logical) + Frames (physical)

✓✓ No external fragmentation

✓✓ Most widely used

(B) Segmentation ★

- Memory logical parts me divided
- Code, stack, data segments

✓✓ Logical view of program

✓✓ Easy protection

(C) Paging + Segmentation (Hybrid)

- Both combined

- Used in advanced systems
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3. Virtual Memory ★ (VERY IMPORTANT)

- Disk space ko RAM jaisa use karta hai
- Allows execution of large programs

Techniques used:

- Paging based virtual memory
 - Swapping
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⚙️ 2. MEMORY ALLOCATION STRATEGIES (IMPORTANT EXTRA)

Ye exam/interview me alag se puch lete hain:

✅ First Fit

- First available block assign

✅ Best Fit

- Smallest suitable block

✅ Worst Fit

- Largest block allocate
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⚠️ 3. FRAGMENTATION (VERY IMPORTANT)

1. Internal Fragmentation

- Fixed block me unused space

2. External Fragmentation

- Free memory scattered hoti hai
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🔄 4. MEMORY MANAGEMENT FUNCTIONS

OS kya-kya karta hai:

- Memory allocation
- Memory deallocation
- Memory protection

- Memory sharing
 - Address mapping
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🧠 FINAL ANSWER TO YOUR QUESTION:

? "Bas 4 techniques hi hoti hai kya?"

👉 ❌ NO

✅ Proper answer:

Memory management includes:

◆ Main Techniques:

- Contiguous Allocation
- Paging
- Segmentation
- Virtual Memory
- (Paging + Segmentation hybrid)

◆ Plus:

- Allocation strategies (First/Best/Worst fit)
 - Fragmentation handling
 - Swapping & address mapping
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🚀 ONE-LINE INTERVIEW ANSWER:

Memory management is not limited to only 4 techniques; it includes allocation strategies, paging, segmentation, virtual memory, and fragmentation handling techniques to efficiently manage RAM.